

Review Session 7

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1 VECM

¹ Let (z_t) be an r -dimensional VAR(p) process given by

$$z_t = \Pi_1 z_{t-1} + \dots + \Pi_p z_{t-p} + \varepsilon_t \quad (1)$$

where $\varepsilon_t \sim I(0)$. We may rewrite our VAR(p) model using the filter

$$\Pi(L) = I - \Pi_1 L - \Pi_2 L^2 - \dots - \Pi_p L^p$$

as

$$\Pi(L)z_t = \varepsilon_t$$

From our discussion in RS1 and RS3, we see that if $\Pi(L)^{-1}$ does exist, it should be given in the form of the following matrix inverse

$$\Pi(z)^{-1} = (I - \Pi_1 z - \Pi_2 z^2 - \dots - \Pi_p z^p)^{-1}$$

Using the same arguments as before, we may also show that if $\Pi(z)^{-1}$ exists in an open set in an open set containing the unit circle, we have the Laurant expansion

$$\Pi(z)^{-1} = \sum_{k=-\infty}^{\infty} \Upsilon_k z^k$$

for some (Υ_k) that are absolutely summable. This means that the filter defined as

$$\Pi(L)^{-1} = \sum_{k=-\infty}^{\infty} \Upsilon_k L^k$$

is well-defined and maps a stationary process to another stationary process as long as $\det(\Pi(z)) \neq 0$ in an open set containing the unit circle. Thus, we see that for (z_t) to

¹Please refer to

1. "Granger's representation theorem: A closed-form expression for I(1) processes", Peter Reinhard Hansen, The Econometrics Journal (2005)
2. "Estimation and Hypothesis Testing of Cointegration Vectors in Gaussian Vector Autoregressive Models", Søren Johansen, Econometrica (1991)

for more details.

be nonstationary, we must have $\det(\Pi(z)) = 0$ for some z such that $|z| = 1$. We are particularly interested in VAR(p) processes that have roots given by $z = 1$.² To this end, we consider a VAR(p) process given in equation (1) such that

$$\Pi(1) = I - \Pi_1 - \Pi_2 - \dots - \Pi_p$$

is not full rank.

We may rewrite equation (1) into the vector error correction form

$$\Delta z_t = -\Pi(1)z_{t-1} + \Gamma_1\Delta z_{t-1} + \dots + \Gamma_{p-1}\Delta z_{t-p+1} + \varepsilon_t$$

where

$$\Gamma_i = -\sum_{j=i+1}^p \Pi_j$$

for $i = 1, \dots, p-1$. To see this, subtract z_{t-1} on both sides of equation (1) and further add and subtract $\left(\sum_{j=i+1}^p \Pi_j\right) z_{t-i}$ for $i = 1, \dots, p-1$ on the right-hand side of equation (1).

From our previous discussion, we see that if $\text{rank}(\Pi(z)) = r$ for all $|z| = 1$, (z_t) has stationary representation and there is no need for the vector error correction form. Thus, we assume that the rank of $\Pi(1)$ is given by $\text{rank}(\Pi(1)) = l < r$. Then, we may decompose $\Pi(1)$ into

$$\Pi(1) = AB'$$

where A and B are both $r \times l$ matrices of full rank. Using this decomposition, we have

$$\Delta z_t = -AB'z_{t-1} + \Gamma_1\Delta z_{t-1} + \dots + \Gamma_{p-1}\Delta z_{t-p+1} + \varepsilon_t \quad (2)$$

The low-rank assumption on $\Pi(1)$ implies that we have at least one unit root so that z_t is not stationary. We want to find the conditions under which the terms $B'z_{t-1}$ and Δz_{t-i} , $i = 1, \dots, p-1$ in (2) become stationary. This is particularly important because the limit distribution of the estimates in the model (2) will become different depending on the order of the regressor and regressand.³ Granger representation theorem in Johansen (1991) gives an answer to this question. Here, we present the theorem as presented by Hansen (2005).

1.1 Granger Representation Theorem

Assumptions Let $\Pi(z) = I - \Pi_1z - \Pi_2z^2 - \dots - \Pi_pz^p$ where $z \in \mathbb{C}$.

1. The roots of the characteristic polynomial (defined from $\det(\Pi(z)) = 0$) are either outside the unit circle or equal to one.
2. The matrix $\Pi(1)$ has reduced rank $l < r$, i.e., we may write $\Pi(1) = AB'$ where A and B are $r \times l$ matrices of full column rank l .

²VAR processes that has roots of modulus one but not $z = 1$ are said to be seasonally integrated or to have seasonal unit roots.

³Note that Δz_t might not be stationary. For example, a VAR(2) process given by $z_t = 2z_{t-1} + z_{t-2} + \varepsilon_t$ implies $\Delta^2 z_t = \varepsilon_t$ so that second differencing is needed to obtain stationarity.

3. The number of unit roots equals $r - l$.

The first assumption ensures that the process is not explosive or seasonally cointegrated. The second assumption ensures that there are at least $r - l$ unit roots and induces cointegration whenever $r \geq 1$. The third assumption restricts the process from being $I(2)$ by restricting the number of unit roots to be exactly $r - l$.

Theorem 1. *Under the assumptions given above, the VECM (2) has the representation*

$$z_t = C \sum_{s=1}^t \varepsilon_s + C(L)\varepsilon_t + C(z_0 - \Gamma_1 z_{-1} - \dots - \Gamma_{p-1} z_{-p+1}) \quad (3)$$

where $C = B_\perp (A'_\perp \Gamma B_\perp)^{-1} A'_\perp$ with $\Gamma = I - \sum_{i=1}^{p-1} \Gamma_i$ and the coefficients of $C(L)$ are given from the recursive formula

$$\Delta C_i = -\Pi(1)C_{i-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta C_{i-j}$$

Proof. Omitted □

From (3), we see that both premultiplying B' to z_t , we get rid of the nonstationary term $C \sum_{s=1}^t \varepsilon_s$ so that $B'z_t$ becomes $I(0)$. Also, taking the first difference of z_t makes the term $C \sum_{s=1}^t \varepsilon_s$ become $C\varepsilon_t$ so that Δz_t is $I(0)$ as well. This implies that z_t is $I(1)$ and that B collects cointegrating vectors of z_t . Furthermore, the nonstationary part of z_t moves only in the direction given by the range of $B_\perp$.

2 Structural Shock Identification

⁴ Assume that the true DGP is given by the SVMA model

$$X_t = \sum_{l=-\infty}^{\infty} \Theta_l \varepsilon_{t-l} \quad (4)$$

where (Θ_l) is square summable and (ε_t) is orthonormal white noise. $\{\varepsilon_t\}$ has the interpretation of being structural shocks. In this case, IRFs are given by just the MA coefficients. We define the impulse responses of X_{it+h} to shock ε_{jt} as

$$\mathbb{E}[X_{it+h} | \varepsilon_{jt} = \delta, \{X_s\}_{s < t}] - \mathbb{E}[X_{it+h} | \varepsilon_{jt} = 0, \{X_s\}_{s < t}] = [\Theta_h]_{ij} \cdot \delta$$

where $[\Theta_h]_{ij}$ is (i, j) -th element of Θ_h .

Remarks:

⁴Please refer to

1. "ABCs(and Ds) of understanding VARs", Fernandez-Villaverde et al (2007)
2. "Recoverability and expectations driven fluctuations", Chahrour and Jurado (2022)
3. "Local projections and VARs estimate the same impulse responses", Plagborg-Moller and Wolf (2021)
4. "Bayesian inference on structural impulse response functions", Plagborg-Moller (2019)

for more details.

1. Imposing a linear VMA model implies that we are identifying the linear transition mechanism of shocks. This means that the value of the benchmark ($\varepsilon_{t-1} = \bar{\varepsilon}$) and the size of the shock ($\varepsilon_t - \varepsilon_{t-1} = \delta$) does not affect the shape of the response.
2. The class of SVMA models encompasses all discrete-time linearized DSGE and SVAR models. On the other hand, SVAR models may not be able to accommodate discrete-time linearized DSGE models, which are given by a state space model.
3. The white noise assumption on the shocks (ε_t) characterizes only the first and second moments of the observables (X_t). Thus, when identifying and estimating the SVMA model, i.e., when identifying and estimating the impulse response functions (Θ_l), the only exploitable information in data comes from the first two moments estimated from the data.

A reduced-form model will give us estimates of model-implied autocovariance matrices. However, autocovariance matrices alone cannot point-identify VMA coefficients. This is because there are an infinite number of (Θ_l) that are compatible with the autocovariance matrices. Even if we know the true $(\Gamma_X(k))_k$, we still cannot point identify (Θ_l). This is the key problem that we want to tackle in identifying structural shocks. Even if we estimate a VMA model directly, we still have the identification problem since there will be many impulse response functions (Θ_l) that give the same likelihood values. In many cases, the VMA model is often too flexible to obtain stable estimates and people use a more restrictive model.

2.1 Identification problem

Let $\Gamma_X := (\Gamma_X(k))_{k=-\infty}^{\infty}$ be the set of autocovariance matrices given by the VMA model (4). We have

$$\begin{aligned}\Gamma_X(k) &= \mathbb{E} \left[\left(\sum_{l=-\infty}^{\infty} \Theta_l \varepsilon_{t-l} \right) \left(\sum_{l=-\infty}^{\infty} \Theta_l \varepsilon_{t-k-l} \right)' \right] \\ &= \sum_{l=-\infty}^{\infty} \sum_{h=-\infty}^{\infty} \Theta_l \Gamma_{\varepsilon}(k+h-l) \Theta_h' \\ &= \sum_{l=-\infty}^{\infty} \Theta_{l+k} \Theta_l'\end{aligned}$$

where we've used the orthonormality of ε_t in the final equality. We may use any reduced-form model to obtain estimates of the autocovariance matrices $\hat{\Gamma}_X = (\hat{\Gamma}_X(k))_{k=-\infty}^{\infty}$.

The identified set of (Θ_l) given by $\hat{\Gamma}_X$ is

$$S(\hat{\Gamma}_X) := \left\{ (\Theta_l) : \sum_{l=-\infty}^{\infty} \Theta_{l+k} \Theta_l' = \hat{\Gamma}_X(k), \quad \forall k = 0, \pm 1, \pm 2, \dots \right\}$$

How do we further characterize $(\Theta_l) \in S(\hat{\Gamma}_X)$? We further point-identify $(\Theta_l) \in S(\hat{\Gamma}_X)$ by placing restrictions on the admissible (Θ_l) justified from economic theory. In following these steps, we have to defend

1. The reduced form model is flexible enough to capture the time variation of the original data while restrictive enough to give stable estimates. These two characteristics are to ensure that $\hat{\Gamma}_X$ is close to Γ_X .
2. The economic theory we use to identify $(\Theta_l) \in S(\hat{\Gamma}_X)$ is justifiable.

After identifying the impulse responses $(\Theta_l) \in S(\hat{\Gamma}_X)$, we may label them in accordance with the economic theory that has been used for identification, e.g., monetary policy shock, new shock, technology shock.

2.2 Structural VAR

Suppose we use a VAR model to obtain the model-implied estimates of the autocovariance matrices. In doing so, people assume

1. The time variations of the observables X_t is well modeled by a VAR(p) model:

$$X_t = \sum_{i=1}^p A_i X_{t-i} + e_t \quad (5)$$

where e_t is an orthogonal white noise process with the variance $\mathbb{E}e_t e_t' = \Sigma$.

2. The structural shocks (ε_t) can be written as linear combination of the projection residuals (e_t) . In other words, we have

$$\begin{aligned} \varepsilon_t &= D[X_t - \mathbb{E}[X_t | \{X_s\}_{s < t}]] \\ &= D e_t \end{aligned}$$

How restrictive is the second assumption? The second assumption implies that the structural shocks (ε_t) must be both invertible and causal with respect to (X_t) . To see this, note that we have invertibility because

$$\varepsilon_t = D[X_t - \mathbb{E}[X_t | \{X_s\}_{s < t}]] \in \text{span}(\{X_t\}_{s \leq t})$$

We also have causality because

$$\begin{aligned} \mathbb{E}[\varepsilon_{t+h} X_t'] &= D \mathbb{E}[(X_{t+h} - \mathbb{E}[X_{t+h} | \{X_s\}_{s < t+h}]) X_t'] \\ &= 0 \end{aligned}$$

for all $h > 0$. We may show the other relation as well. Suppose that the structural shocks (ε_t) are both invertible and causal with respect to (X_t) . Then, we have

$$\begin{aligned} e_t &= X_t - \mathbb{E}[X_t | \{X_s\}_{s < t}] \\ &= \sum_{l=0}^{\infty} \Theta_l \varepsilon_{t-l} - \mathbb{E} \left[\sum_{l=0}^{\infty} \Theta_l \varepsilon_{t-l} \middle| \{X_s\}_{s < t} \right] \\ &= \sum_{l=0}^{\infty} \Theta_l \varepsilon_{t-l} - \mathbb{E} \left[\sum_{l=0}^{\infty} \Theta_l \varepsilon_{t-l} \middle| \{\varepsilon_s\}_{s < t} \right] \\ &= \Theta_0 \varepsilon_t \end{aligned}$$

where the second and third equality comes from causality and invertibility of (ε_t) respectively. Thus, we see that under both causality and invertibility, the true structural shocks ε_t are given by $\varepsilon_t = \Theta_0^{-1}e_t$. Here, Θ_0 is often called the at-impact response matrix. WLOG, we assume a VAR(1) model because, for any VAR(p), we may construct a companion-form VAR(1). We call the model

$$X_t = AX_{t-1} + e_t$$

the reduced form VAR, and the model

$$X_t = AX_{t-1} + \Theta_0\varepsilon_t$$

structural form VAR⁵. Note that we have

$$X_t = \Theta_0\varepsilon_t + A\Theta_0\varepsilon_{t-1} + A^2\Theta_0\varepsilon_{t-2} + \dots$$

Since A can be estimated from the model, we may obtain the full set of impulse responses $\Theta_l = A^l\Theta_0$ if we identify Θ_0 . Not all Θ_0 are compatible with the given VAR(1) model. Notice that we have

$$\Sigma = \mathbb{E}e_te_t' = \mathbb{E}\Theta_0\varepsilon_t\varepsilon_t'\Theta_0' = \Theta_0\Theta_0'$$

Thus, we identify Θ_0 up to a rotation matrix R such that $\Theta_0 = \tilde{\Theta}_0R$ where $\tilde{\Theta}_0$ is any matrix with $\tilde{\Theta}_0\tilde{\Theta}_0' = \Sigma$. Without loss of generality, we may assume that $\tilde{\Theta}_0$ comes from the Cholesky decomposition of Σ .

2.3 Relationship between IRF and VD

Consider the structural form of VAR

$$X_t = AX_{t-1} + \tilde{\Theta}_0R\varepsilon_t$$

The impulse responses may be obtained after recursive substitution

$$X_t = \tilde{\Theta}_0R\varepsilon_t + A\tilde{\Theta}_0R\varepsilon_{t-1} + A^2\tilde{\Theta}_0R\varepsilon_{t-2} + \dots$$

Thus, we have

$$\Theta_h = A^h\tilde{\Theta}_0R$$

as the IRF at horizon h where $\tilde{\Theta}_0R$ is the at impact response matrix. Note that the (i, j) -th element of Θ_h is the impulse response of the i -th variable to j -th shock. The forecast error and forecast error variance are given by

$$\begin{aligned} X_{t+h} - \mathbb{E}[X_{t+h}|\{X_s\}_{s \leq t}] &= \tilde{\Theta}_0R\varepsilon_{t+h} + A\tilde{\Theta}_0R\varepsilon_{t+h-1} + \dots + A^{h-1}\tilde{\Theta}_0R\varepsilon_{t+1} \\ &= \Theta_0\varepsilon_{t+h} + \Theta_1\varepsilon_{t+h-1} + \dots + \Theta_{h-1}\varepsilon_{t+1} \end{aligned}$$

⁵Sometimes people write

$$\Theta_0^{-1}X_t = \Theta_0^{-1}AX_{t-1} + \varepsilon_t$$

to mean the structural form VAR.

and

$$\text{Var}(X_{t+h} - \mathbb{E}[X_{t+h} | \{X_s\}_{s \leq t}]) = \sum_{l=0}^{h-1} \Theta_l \Theta_l'$$

so that the forecast error variance is just the sum of the outer product of the IRFs. Furthermore, recall that we can interpret matrix multiplication as the sum of column $\times$ row products. I.e., let

$$\Theta_l = (\theta_{1l} \quad \dots \quad \theta_{nl})$$

where each column $\theta_{jl} \in \mathbb{R}^n$ collects the responses to j -th shock at horizon l . Then, we have

$$\Theta_l \Theta_l' = \sum_{j=1}^n \theta_{jl} \theta_{jl}'$$

Thus, we can write VD as

$$\sum_{l=0}^{h-1} \Theta_l \Theta_l' = \sum_{l=0}^{h-1} \sum_{j=1}^n \theta_{jl} \theta_{jl}' = \sum_{j=1}^n \left(\sum_{l=0}^{h-1} \theta_{jl} \theta_{jl}' \right)$$

Let $X_t = \begin{pmatrix} X_{1t} \\ \vdots \\ X_{nt} \end{pmatrix}$. How do we decompose the forecast error variance of X_{it} to variances explained by each of the identified shocks?

1. The (i, i) -th element of $\sum_{l=0}^{h-1} \Theta_l \Theta_l'$ is forecast error variance of X_{it} .
2. (i, i) -th element of $\sum_{l=0}^{h-1} \Theta_{jl} \Theta_{jl}'$ is the component of forecast error variance of X_{it} that is explained by j -th structural shock.

We can also think of forecast error covariance of X_{pt} and X_{qt} .

1. The (p, q) -th element of $\sum_{l=0}^{h-1} \Theta_l \Theta_l'$ is forecast error covariance of X_{pt} and X_{qt} .
2. The (p, q) -th element of $\sum_{l=0}^{h-1} \Theta_{jl} \Theta_{jl}'$ is the component of forecast error covariance of X_{pt} and X_{qt} that is explained by j -th structural shock.